U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL AVIATION ADMINISTRATION

AIR TRAFFIC ORGANIZATION POLICY

ORDER
JO 6130.3D

Effective Date:
xx/xx/2019
Implementation Date:
xx/xx/2019

SUBJ: Maintenance of Flight Data Input/Output (FDIO) Equipment

This order prescribes technical standards, tolerances, maintenance, and certification requirements for the Flight Data Input/Output (FDIO) system. This order provides the necessary guidance, to be used in conjunction with information available in instruction books, for the proper maintenance of the National Airspace System (NAS).

This revision cancels Order JO 6130.3C, Maintenance of Flight Data Input/Output (FDIO) Equipment Change (CHG) 2, dated 02/07/2016. The changes are a result of Configuration Control Decision (CCD)-L36760, ZHN ARTCC FDIO-G Hardware Installation, and CCD-N36903, Personal Computer-Remote Control Unit (PC-RCU) Software Release Version 3.3.0 and Printer Replacement.

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RECORD OF CHANGES DIRECTIVE NO. JO 6130.3D

CHANGE TO BASIC	MM	DD	YY	OPTIONAL	CHANGE TO BASIC	MM	DD	YY	OPTIONAL
LOA (ACY)	03	28	2018						
LOA (ZSU)	05	30	2018						
LOA (OGG)	07	30	2018						
LOA (ANC, ANCA)	09	18	2018						
LOA (MRI)	10	25	2018						
LOA (TIK)	11	26	2018						
LOA (All sites)	12	14	2018						
LOA (All sites)	06	24	2019						

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Chapter 1. General Information and Requirements

1–1. Purpose**.** This Maintenance Technical Handbook (MTHB) provides guidance and prescribes technical standards, tolerances, and procedures applicable to the maintenance and inspection of Flight Data Input/Output (FDIO) equipment. It also provides information on special methods and techniques, which will enable maintenance personnel to achieve optimum performance from the equipment. This information augments information available in Technical Instruction (TI) books and other MTHBs, and complements the latest edition of Order 6000.15, General Maintenance Handbook for National Airspace (NAS) Facilities**.**

1**–**2. Audience**.** This MTHB requires actions by the Airway Transportation System Specialist (ATSS) at operational facilities with Facility, Service, and Equipment Profile (FSEP) equipment: FDIO Center (FDIOC) and FDIO Remote (FDIOR).

1**–**3. Where Can I Find This Order? An electronic version of this MTHB can be found at https://employees.faa.gov/tools_resources/orders_notices/.

1**–**4. Cancellations. This order cancels Order JO 6130.3C, Maintenance of the Flight Data Input/Output (FDIO) Equipment, dated May 9, 2014.

1–5. Explanation of Changes. Updates reflect changes resulting from the following:

a. Configuration Control Decision (CCD)-L36760, ZHN ARTCC FDIO-G Hardware Installation, and SSM-FDIO-065, FDIO-G Hardware/Software Installation at Honolulu Control Facility (HCF). SSM-FDIO-065 cut over terminal local and remote sites from the Series/1 Replacement (S1R) to the FDIO-Gateway (FDIO-G) via the Offshore Flight Data Processing System (OFDPS) Communication Gateway (OCG).

b. CCD-N36903, Personal Computer-Remote Control Unit (PC-RCU) Software Release Version 3.3.0 and Printer Replacement, and SSM-FDIO-067, PC-RCU Software Release Version 3.3.0 and Printer Replacement. SSM-FDIO-067 replaces the IER Replacement Flight Strip Printer (RFSP) with the Boca Replacement Flight Strip Printer (RFSP).

c. CCD-L36902, Moving HCF Connected PC-RCU's to TI 6130.6 Baseline, and SSM-FDIO-069, Updating HCF Connected PC-RCUs. Changes were made to distinguish which TI manual to reference for PC-RCU equipment. In prior issuances of this document, PC-RCU sites in the CONUS and ZAN environments referenced TI 6130.6 and HCF, and HCF connected PC-RCU sites referenced TI 6130.20. Upon completion of SSM-FDIO-069, HCF and HCF connected remote sites will transition from the legacy PC-RCU software (version 2.3) and TI 6130.20 to the new Linux based software (version 3.x.x) and TI 6130.6. PC-RCU sites will now reference the appropriate PC-RCU TI manual based on the PC-RCU software they are running and not their location. For CONUS and ZAN PC-RCU sites this does not change your applicable TI (TI 6130.6), but for HCF and HCF connected remote sites, this allows them to reference the proper TI as they transition to the new software version. Once all sites in the HCF environment are transitioned, TI 6130.20 will be decommissioned and references to it will be removed from this document.

d. CCD-P37685, Implementation of New FDIO Requirements In Support of the Terminal Flight Data manager (TFDM) Program at Phoenix ATCT & TRACON, and SSM-FDIO-076, FDIO PC-RCU Software Version PX400 PHX Only. Changes were made in this revision to add

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references to TI 6130.22, Flight Data Input/Output (FDIO) Personal Computer-Remote Control Unit (PC-RCU) for sites with Flight Data Transfer Service (FDTS). FDIO-R sites authorized to enable Terminal Flight Data Manager (TFDM) Service and/or Flight Data Transfer Service (FDTS) (running PC-RCU software version 4.x.x) will now reference TI 6130.22. FDIO-R sites (running version 3.x.x) will continue to reference TI 6130.6. Also, references to TI 6130.20 have been removed since HCF and all HCF connected remote sites have transitioned to PC-RCU software version 3.x.x, and now use TI 6130.6.

1–6. Certification. Refer to Order 6000.15 for general guidance on the certification of systems, subsystems, and equipment. Refer to Appendix 1, Certification, of this MTHB for specific FDIO certification requirements.

1–7. Aircraft Accident. When aircraft accidents or incidents occur, Air Traffic Organization Technical Operations personnel are responsible, when requested by the Technical Operations Aircraft Accident Representative (TOAAR) through the appropriate control center, to evaluate and document the technical performance of the facilities which may have been involved (for some facilities, it may also be necessary to remove them from service, and to conduct flight inspections). This requires that facility operational data be obtained and recorded in the maintenance log and on technical performance records. These records are official documents, and may be used by an aircraft accident investigation board in the determination of facility operational status at the time of the accident. Refer to the latest edition of Order JO 8020.16, Air Traffic Organization Aircraft Accident and Incident Notification, Investigation, and Reporting, for detailed guidance on requirements and activities.

1–8. Maintenance and Modification Procedure. Order 6000.15, this MTHB, the applicable equipment TIs, and other applicable handbooks shall be consulted and used together by the maintenance technician in all duties and activities for the maintenance of FDIO. These documents are considered collectively as the single official source of maintenance policy and direction authorized by Technical Operations Services. References located in the appropriate paragraphs of this handbook entitled Chapter 3, Standards and Tolerances, Chapter 4, Maintenance Requirements, and Chapter 5, Maintenance Procedures, indicate to the user whether this handbook and/or the equipment TI must be consulted for a particular standard, key inspection element or performance parameter, performance check, maintenance task, or maintenance procedure.

1–9. Risks.

a. Operational. In compliance with the latest edition of Order JO 6000.50, Technical Operations National Airspace System (NAS) Integrated Risk Management, activities are to be assessed for potential risk to the NAS from an operational perspective. **There are no known operational risks associated with this MTHB.**

b. Safety. In compliance with the latest editions of Order 1100.161, Air Traffic Safety Oversight, and JO 1000.37, ATO Safety Management System, all NAS changes, including MTHBs, require a Safety Risk Management (SRM) assessment prior to delivery. The SRM information for this MTHB is attached as Appendix 7, Safety Risk Management Documents, and they are available at **https://ksn2.faa.gov/aje/aje3/safety/default.aspx.** For further guidance in developing SRM documentation, refer to the latest version of the Safety Management System (SMS) Manual. **There are no known safety risks associated with this MTHB.**

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c. Security. In compliance with the latest edition .of Order 1370.121, FAA Information Security and Privacy Program & Policy, the FAA must ensure that security controls are implemented and commensurate with the risk and magnitude of the harm that would result from the loss, misuse, denial of service, unauthorized access, or modification of Federal information assets. **There are no known security risks associated with this MTHB.**

1–10. Implementation Date**.** This MTHB is applicable to all FDIOC and FDIOR sites upon issuance.

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Chapter 2. Technical Characteristics

2–1. Purpose or Function**.** FDIO provides input and output routing of flight plan data, weather information, and general information between central Air Traffic Control (ATC) computer systems and Air Traffic Controllers (controllers) located at approximately 700 remote sites. With the exception of the Honolulu Control Facility (HCF), FDIO does not provide services to controllers located within an Air Route Traffic Control Center (ARTCC).

From an FDIO perspective, central ATC computer systems include the En Route Automation System (EAS), Micro En Route Automated Radar Tracking System/Flight Data Processor-2000 (MEARTS/FDP–2000), and the MEARTS/Offshore Flight Data Processing System (OFDPS). An FDIO remote site is any Air Traffic Control Tower (ATCT), Radar Approach Control (RAPCON), Center Radar Approach Control (CERAP), or Terminal Radar Control (TRACON) facility having FDIO equipment.

2–2. System Description**.** FDIO consists of two major subsystems:

1. FDIO Gateway (FDIO–G) located at 20 Contiguous US (CONUS) ARTCCs, the Anchorage ARTCC (ZAN), and the Honolulu ARTCC (ZHN) co-located at HCF.

2. Personal Computer–Remote Control Units (PC–RCU) located at nearly 700 remote sites and the HCF. Communications between FDIO-G and the PC–RCU is primarily enabled using individual FAA Telecommunications Infrastructure (FTI) serial circuits. A small number of remote sites that are not supported by FTI continue to employ various legacy communication links to facilitate FDIO services. These legacy links include Bandwidth Manager (BWM), the Alaska NAS Interfacility Communications System (ANICS), and the FAA Data Multiplexing Network (DMN).

Located in an ARTCC, FDIO–G is a router-based subsystem that essentially communicates with the center's central ATC computer (e.g., EAS) on one side via the En Route Communications Gateway (ECG), and with remotely located PC–RCUs on the other side. Figure 2–1, General CONUS FDIO Architecture and Interfaces, depicts the general domestic system architecture.

ZAN and HCF are configured in somewhat different fashions; ZAN and HCF do not employ an ECG. Instead, the ZAN central ATC computer, FDP–2000, interfaces directly with FDIO-G to the remotely located PC-RCUs, and the HCF central ATC computer (OFDPS) communicates through the OFDPS OCG and FDIO–G to both local and remote PC-RCUs. Figure 2–2, General ZAN FDIO Architecture and Interfaces, depicts the system architecture at the Anchorage ARTCC, and Figure 2–3, General HCF FDIO Architecture and Interfaces, depicts the configuration at the HCF.

Located in the Terminal environment (ATCT, RAPCON, CERAP, or TRACON), the PC–RCU provides the controller interface for FDIO-supported flight, weather, and general data. The PC–RCU constitutes the remote interface to the center's FDIO–G or HCF OCG equipment. The PC–RCU drives local FDIO printers, monitors, and keyboards. The PC–RCU also provides a flight data feed to other systems external to FDIO via the echo port.

The Offshore Terminal environment of the HCF is unique. At the HCF, FDIO–G communicates with OFDPS on one side via the OCG, with both local and remote PC–RCUs. Within the HCF,

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FDIO printers are used, but due to space limitations, the FDIO keyboards and monitors are emulated by MEARTS. And, because the displays in the Offshore environment are Video Graphic Array (VGA) displays, a video display terminal interface device is employed between the HCF local PC-RCU and the local VGA monitor to perform the required RS-422 to VGA conversion.

The San Juan CERAP is another unique FDIO environment. The CERAP PC-RCUs drive their local FDIO peripherals using En Route-formatted strips, rather than the usual Terminal-formatted strips, yet the CERAP PC–RCUs are driven as a standard remote PC–RCU via the Miami ARTCC's (ZMA) FDIO–G. This configuration is enabled by the use of a local ZMA Host patch to support En Route FDIO services to and from the CERAP.

To summarize, the following FDIO equipment is used in these distinct environments:

a. En Route centers:

FDIO–G
Peripherals: None

b. Terminal:

PC–RCU
Peripherals: keyboards, monitors, and printers

c. Offshore (HCF):

FDIO–G

PC–RCU
Peripherals: printers, video display terminal interface adapter

d. CERAP (San Juan):

PC–RCU
Peripherals: keyboards, monitors, and printers

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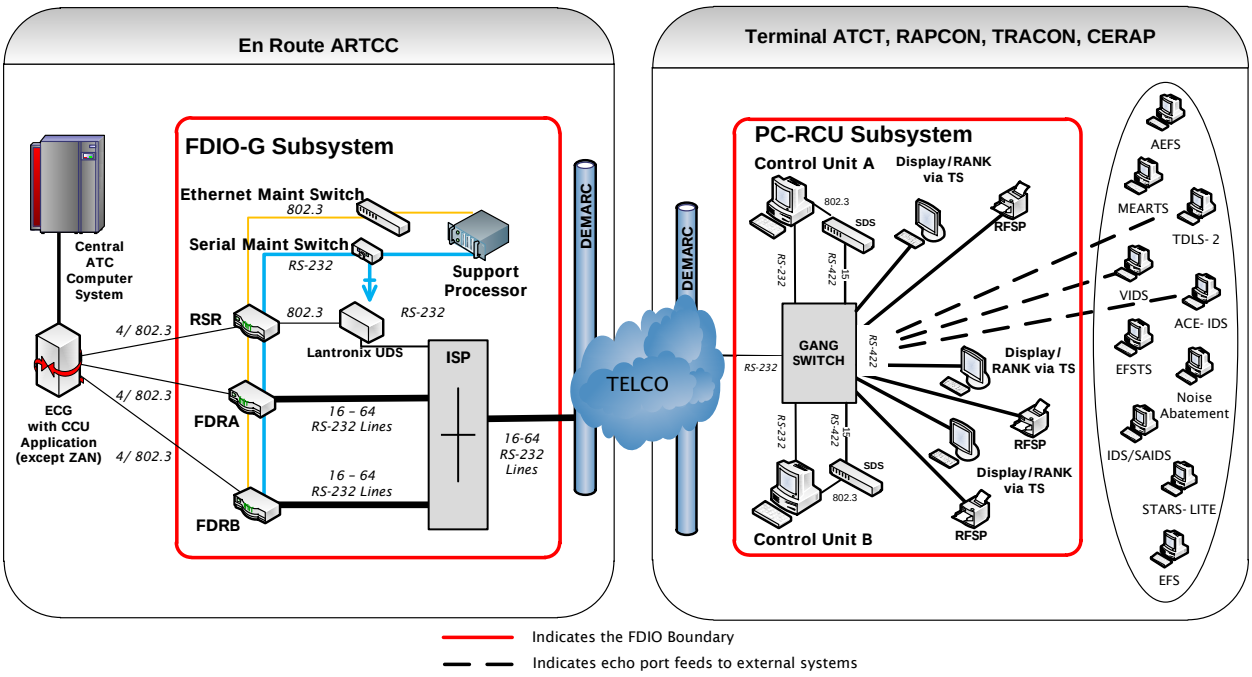

Figure 2–1. General CONUS FDIO Architecture and Interfaces

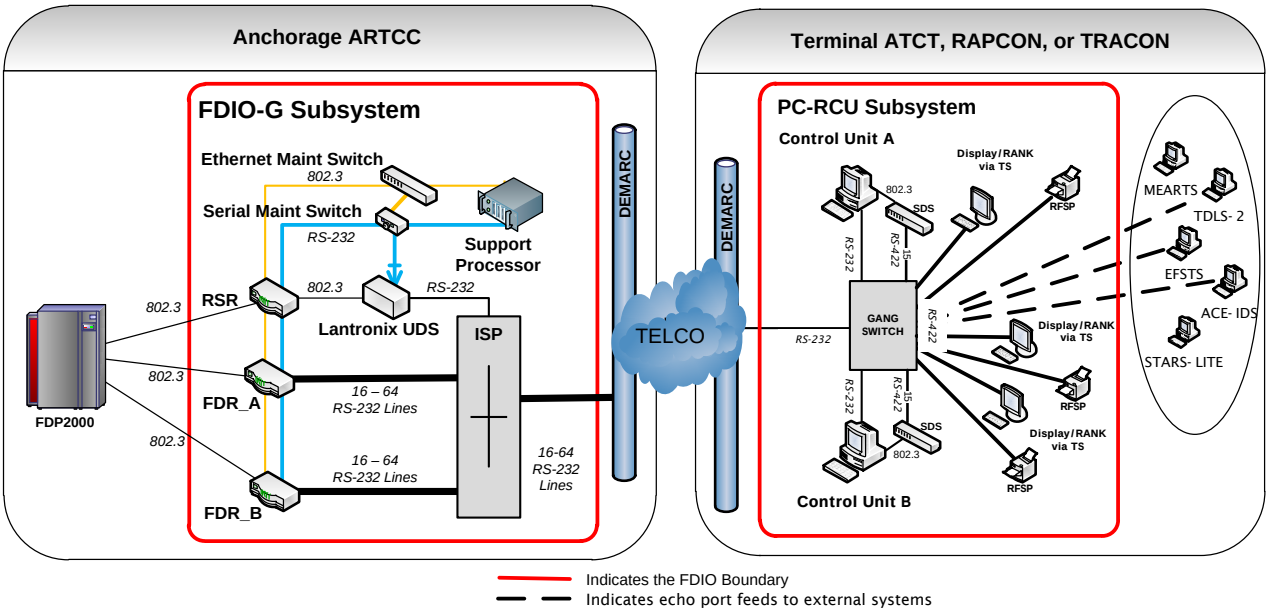

Figure 2–2. General ZAN FDIO Architecture and Interfaces

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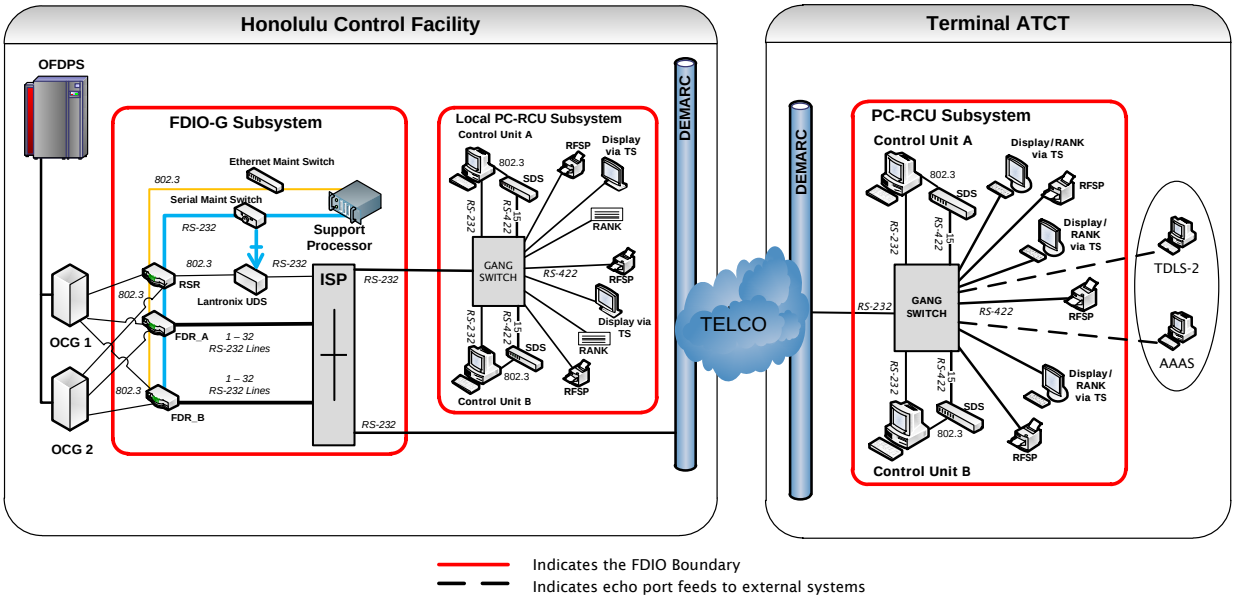

Figure 2–3. General HCF FDIO Architecture and Interfaces

2-3. Theory**.** Technical descriptions and theory of operation are contained in the following FDIO TI manuals:

a. For FDIO–G: TI 6130.16, En Route Flight Data Input/Output (FDIO) System Hardware Design Document (HDD) Instruction Book: FDIO-Gateway (FDIO–G).

b. For PC–RCU sites running software version 3.x.x: TI 6130.6.

c. For PC–RCU sites running software version 4.x.x: TI 6130.22.

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Chapter 3. Standards and Tolerances

3–1. General. This chapter prescribes the standards and tolerances for FDIO equipment, as defined and described in Order 6000.15. All key performance parameters and/or key inspection elements are clearly identified by an arrow (→) placed to the left of the applicable item.

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STANDARDS AND TOLERANCES

Parameter	*Reference Paragraph*	*Standard*	*Tolerance/Limit*	
			Initial	*Operating*
3–2. FDIO.				
→ **a.** ARTCC				
(1) CONUS ……	Par. 5–5	Satisfactory input, transfer, and output of flight data from EAS/FDP–2000 to remote site, including correct device response, as verified using online or offline certification program, or NAS Test Device (TD) message.	None	None
(2) ZAN …….	Par. 5–7			
b. HCF.				
→ **(1)** From OFDPS ….. to remote site.	Par. 5–6	Satisfactory input, transfer and output of flight data from OFDPS to remote site, including correct device response, as verified using online or offline certification program, or NAS TD message.	None	None
→ **(2)** From OFDPS ...… to HCF-located FDIO position.	Par. 5–6	Satisfactory input, transfer, and output of flight data from OFDPS to HCF-located FDIO position, including correct device response, as verified using online or offline certification program, or NAS TD message.	None	None
→ **(3)** From HCF-…….. located FDIO keyboard to OFDPS.	Par. 5–6	Satisfactory input, transfer, and output of flight data from the HCF-located FDIO keyboard to the OFDPS, including correct device response, as verified using NAS TD message.	None	None
→ **c.** Terminal.	Par. 5–5	Satisfactory input, transfer, and output of flight data from the terminal FDIO position to the EAS/FDP–2000/OFDPS, including correct device response, registration and print quality as verified using NAS TD message**.**	None	None

STANDARDS AND TOLERANCES (Continued)

Parameter	*Reference Paragraph*	*Standard*	*Tolerance/Limit*	
			Initial	*Operating*
3–3. FDIO–G.				
a. Run SP Power-On …..… Self-Test (POST).	TI 6130.16 Par. 6.1.1	Normal front and rear panel indications after POST	Same as Standard	Same as Standard
b. Run FDR POST ……	TI 6130.16 Par. 6.1.2	Normal front and rear panel indications after POST	Same as Standard	Same as Standard
c. Run RSR POST …….	TI 6130.16 Par. 6.1.3	Normal front and rear panel indications after POST	Same as Standard	Same as Standard
d. Switch FDR ……...	TI 6130.16 Par. 6.1.6 and Par. 6.1.7	Error free	Same as Standard	Same as Standard
e. Verify FDR and RSR … to ECG/FDP–2000/OCG redundant path.	TI 6130.16 Par. 6.1.4 and Par. 6.1.5	Error free	Same as Standard	Same as Standard
f. Review system logs …...	TI 6130.16 Par. 6.1.8	No evidence of problems or errors.	Same as Standard	Same as Standard
g. Archive system logs …..	TI 6130.16 Par. 6.2.1	Error free completion	Same as Standard	Same as Standard
h. Inspect equipment …….	TI 6130.16 Par. 6.2.2	No ventilation obstructions. No damage or tampering. No unauthorized connections. No unauthorized modifications. All properly authorized connections are in place.	Same as Standard	Same as Standard
i. ISP Control Card ……... battery check.	TI 6130.16 Par. 6.2.4	> 0.8 V	Same as standard	Same as standard

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STANDARDS AND TOLERANCES (Continued)

Parameter	*Reference Paragraph*	*Standard*	*Tolerance/Limit*	
			Initial	*Operating*
3–4. PC–RCU (Software v. 3.x.x).				
a. Switch control units	TI 6130.6 Par. 6.1.1.1	Error free	Same as Standard	Same as Standard
b. Run POST …….	TI 6130.6 Par. 6.1.1.2	Error free completion	Same as Standard	Same as Standard
3–5. PC–RCU (Software v. 4.x.x)				
a. Switch control units ….	TI 6130.22 Par. 6.1.1.1	Error free	Same as Standard	Same as Standard
b. Run POST …….	TI 6130.22 Par. 6.1.1.2	Error free completion	Same as Standard	Same as Standard

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Chapter 4. Maintenance Requirements

4–1. General. This chapter establishes all the maintenance activities that are required for FDIO equipment on a periodic, recurring basis, and the schedules for their accomplishment. The chapter is divided into two sections. The first section identifies the performance checks (tests, measurements, and observations) of normal operation controls and functions, which are necessary to determine whether operation is within established tolerances/limits. The second section identifies other tasks that are necessary to prevent deterioration and ensure reliable operation. An safety-related checks with direct impact to safety of flight within the NAS are clearly identified by a pound sign (#) placed to the left of the applicable task.

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Section 1. PERFORMANCE CHECKS

Performance Checks	*Reference Paragraph*	
	Standards & Tolerances	*Maintenance Procedures*
4–2. FDIO.		
a. Weekly.		
(1) ARTCC.		
Satisfactory input, transfer and output of …..... flight data to remote site as determined at the ARTCC using online or offline certification programs, or NAS TD message.	CONUS Par. 3–2.a.(1) ZAN Par. 3–2.a.(2)	Par. 5–5 Par. 5–7
(2) HCF.		
(a) Satisfactory input, transfer, and …….. output of flight data to remote site as determined at the HCF using online or offline certification programs, or NAS TD message.	Par. 3–2.b.(1)	Par. 5–6
(b) Satisfactory input, transfer, and …….. output of flight data to HCF-located FDIO position using online or offline certification programs, or NAS TD message.	Par. 3–2.b.(2)	Par. 5–6
b. Monthly.		
(1) HCF.		
Satisfactory input, transfer, and output ……. of flight data from the HCF-located FDIO keyboard to the OFDPS using NAS TD message.	Par. 3–2.b.(3)	Par. 5–6
(2) Terminal.		
Correct device response, registration, …….. printing and entry of data as determined at the terminal facility using NAS TD message.	Par. 3–2.c.	Par. 5–5

Section 1. PERFORMANCE CHECKS (Continued)

Performance Checks	*Reference Paragraph*	
	Standards & Tolerances	*Maintenance Procedures*
4–3. FDIO–G.		
a. Daily.		
Review system logs ……	Par. 3–3.f.	TI 6130.16, Par. 6.1.8
b. Monthly . ……...		
Switch FDR ……...	Par. 3–3.d.	TI 6130.16, Par. 6.1.6 and Par. 6.17
c. Quarterly.		
(1) Run SP POST …….……	Par. 3–3.a.	TI 6130.16, Par. 6.1.1
(2) Run FDR POST …….…….……	Par. 3–3.b.	TI 6130.16, Par. 6.1.2
(3) Run RSR POST ……..……	Par. 3–3.c.	TI 6130.16, Par. 6.1.3
(4) Verify FDR and RSR to ..... ECG/FDP–2000/OCG redundant path.	Par. 3–3.e.	TI 6130.16, Par. 6.1.4 and Par. 6.1.5
4–4. PC–RCU.		
a. Monthly.		
Switch control units …….……..……...	PC-RCU (software v. 3.x.x): Par. 3–4.a.	TI 6130.6, Par. 6.1.1.1
	PC-RCU (software v. 4.x.x): Par. 3–5.a.	TI 6130.22, Par. 6.1.1.1
b. Quarterly.		
Run POST on control units ……..	PC-RCU (software v. 3.x.x): Par 3–4.b.	TI 6130.6, Par. 6.1.1.2
	PC-RCU (software v. 4.x.x): Par. 3–5.b.	TI 6130.22, Par. 6.1.1.2

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Section 2. OTHER MAINTENANCE CHECKS

Maintenance Checks	*Reference Paragraph*	
	Standards & Tolerances	*Maintenance Procedures*
4–5. FDIO–G.		
a. Monthly.		
(1) Archive system logs ……	Par. 3–3.g.	TI 6130.16, Par. 6.2.1
(2) Inspect equipment ……	Par. 3–3.h.	TI 6130.16, Par. 6.2.2
b. Quarterly.		
Review system accounts …….	None	TI 6130.16, Par. 6.2.3
c. Annually.		
(1) ISP — Check control card battery ……	Par. 3–3.i.	TI 6130.16, Par. 6.2.4
(2) SP — Clean exterior …….	None	TI 6130.16, Par. 6.2.5
(3) FDR — Clean exterior …….	None	TI 6130.16, Par. 6.2.5
(4) RSR — Clean exterior ……..	None	TI 6130.16, Par. 6.2.5
(5) ISP — Clean exterior ……	None	TI 6130.16, Par. 6.2.5

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Section 2. OTHER MAINTENANCE CHECKS (Continued)

Maintenance Checks	*Reference Paragraph*	
	Standards & Tolerances	*Maintenance Procedures*
4–6. PC–RCU.		
a. Monthly.		
(1) Replacement Flight Strip Printer (RFSP) …... (IER or Boca).		
Clean print head and platen ……..	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 6.1.2.9.1; PC-RCU (software v. 4.x.x): TI 6130.22, Par. 6.1.2.9.1
(2) Replacement Alphanumeric Keyboard …..... (RANK).		
(a) Clean exterior ……	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 6.1.2.8; PC-RCU (software v. 4.x.x): TI 6130.22, Par. 6.1.2.8
(b) Verify lamp brightness and control …...	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 3.2.6; PC-RCU (software v. 4.x.x): TI 6130.22, Par. 3.2.6
(3) Display.		
(a) Clean exterior and screen. …….....	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 6.1.2.6; PC-RCU (software v. 4.x.x.): TI 6130.6, Par. 6.1.2.6
(b) Verify contrast and brightness. ……..	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 3.2.7; PC-RCU (software v. 4.x.x): TI 6130.22, Par. 3.2.7

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Section 2. OTHER MAINTENANCE CHECKS (Continued)

Maintenance Checks	*Reference Paragraph*	
	Standards & Tolerances	*Maintenance Procedures*
4–6. PC–RCU (Continued)		
(4) Terminal Server.		
Clean exterior ..……..	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 6.1.2.7: PC-RCU (software v. 4.x.x): TI 6130.22, Par. 6.1.2.7
b. Quarterly.		
Verify fan operation in control units. ……..……...	None	None
c. Semiannually.		
(1) RFSP (IER).		
(a) Clean cutter blade, blade down ……. sensor and interior.	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 6.1.2.9.4.1; PC-RCU (software v. 4.x.x): TI 6130.22, Par. 6.1.2.9.4.1
NOTE: Semiannually or as indicated by the printer flashing "Service" on the front display panel.		
(b) Platen roller brush cleaning. …..…….	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 6.1.2.9.4.3; PC-RCU (software v. 4.x.x.): TI 6130.22, Par. 6.1.2.9.4.3

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Section 2. OTHER MAINTENANCE CHECKS (Continued)

Maintenance Checks	*Reference Paragraph*	
	Standards & Tolerances	*Maintenance Procedures*
(2) RFSP (Boca).		
Clean Cut Opto sensor, Load SQ sensor, …..... and Cutter Assembly.	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 6.1.2.9.4.2; PC-RCU (software v. 4.x.x.): TI 6130.22, Par. 6.1.2.9.4.2
d. Annually.		
System Console.		
Clean exterior. ..……..……..	None	PC-RCU (software v. 3.x.x): TI 6130.6, Par. 6.1.2.5: PC-RCU (software v. 4.x.x): TI 6130.22, Par. 6.1.2.5

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Chapter 5. Maintenance Procedures

5–1. General. This chapter establishes the procedures for accomplishing the various essential maintenance activities, which are required for FDIO equipment, on either a periodic or an incidental basis. The chapter is divided into three sections. The first section describes the procedures to be used in making the performance checks listed in chapter 4, section 1. The second section describes the procedures for doing the tasks listed in chapter 4, section 2. The third section describes procedures for doing special tasks, usually nonscheduled and not listed in chapter 4.

5–2. FAA Forms 6000 Series. Order 6000.15 contains guidance and detailed instructions for field utilization of FAA Form 6000 series (Trend Analysis), as applicable to the FDIO. Make entries in accordance with the instructions published in Order 6000.15, (except as otherwise instructed in the subparagraphs to follow). Forms are available at https://employees.faa.gov/tools_resources/forms/.

5–3. Remote Maintenance Monitoring (RMM). Technical Operations will identify maintenance activities that can be accomplished via RMM. They include:

- Monitoring system status and alarms
- Managing system configuration
- Performing fault isolation and restoration
- Conducting analysis of system performance
- Performing periodic maintenance (if applicable)
- Performing certification (if applicable)

More information on RMM is provided in the latest editions of Order 6000.30 and Order 6000.15.

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Section 1. PERFORMANCE CHECK PROCEDURES

5–4. General. All performance check procedures other than those detailed in this section are contained in the following TIs:

a. FDIO–G: TI 6130.16

b. PC–RCU sites running software version 3.x.x: TI 6130.6

c. PC–RCU sites running software version 4.x.x: TI 6130.22.

5–5. Test Device Message Procedure for the CONUS ARTCC or Terminal Facility

a. Object. Verify FDIO operation between the central ATC computer and a terminal facility. This will test the FDIO interface and verify the correct operation of the printer.

b. **Discussion.** This procedure will cause a TD message to be sent to the requested FDIO printer.

c. **Test Equipment Required.** None.

d. Conditions. Systems must be operational and interfaced.

e. Detailed Procedure.

From a CONUS ARTCC, the following is executed from an operational EAS display.

From a Terminal facility, the following is executed from an operational FDIO keyboard.

Enter the following TD command for the position being tested:

TD aaaL or TD aaaLbbb

(1) Where aaa is the identifier for the tower or approach control facility.

(2) Where L is replaced with T for tower printer, D for approach control departure printer, A for approach control arrival printer, or O for approach control overflight printer. (L must be either T, D, A, or O.)

(3) Where bbb is required only if a printer name is adapted for an approach control departure or arrival printer.

NOTE: Refer to NAS–MD–311, Computer Program Functional Specification Message Entry and Checking, Paragraph 8.5, Test Device, for complete format.

5–6. TD Procedure for HCF and HCF-Connected PC-RCUs with Local and Remote FDIO Printers and CRTs.

a. Object. Verify local FDIO operation within the HCF. This will test the FDIO interface and verify the correct operation of the printer or CRT.

b. Discussion. The TD message is sent from an HCF-located FDIO keyboard to OFDPS, and is then routed back to either the printer or CRT associated with that keyboard, depending on the command used below.

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c. Test Equipment Required. None.

d. Conditions. Systems must be operational and interfaced.

e. Detailed Procedure.

From an operational FDIO keyboard located within the HCF, enter the following TD command for the peripheral being tested:

TD P or TD

(1) Where P will cause the TD message to print on the printer associated with the keyboard.

(2) If the P does not follow the TD, the TD message will be displayed on the CRT associated with the keyboard.

NOTE: Refer to NAS–MD–5311, Message Entry and Checking, Paragraph 6.2, Test Device Message, for complete format.

5–7. TD Procedure for ZAN to Terminal Facility**.**

a. Object. Verify FDIO operation. This will test the interface and verify the correct operation of the FDIO interface and printer.

b. Discussion. This procedure will cause a TD message to be sent to the requested terminal printer.

c. Test Equipment Required. None.

d. Conditions. Systems must be operational and interfaced.

e. Detailed Procedure.

From an operational FDP–2000 KVDT, enter the following TD command for the terminal printer being tested:

TD sss

(1) Where sss is the sector number of the printer where the message will be printed.

(2) If the sector number is omitted, and the entering source is not a supervisory position, the TD message will be printed on the printer associated with the keyboard from which the TD was entered.

(3) If the TD is entered at a supervisory station and the sector number is omitted, the TD message will be printed on all printers.

5–8. – 5–49. Reserved**.**

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Section 2. OTHER MAINTENANCE TASK PROCEDURES

5–50. General**.** All other maintenance task procedures for the FDIO–G are contained in TI 6130.16.

All other maintenance task procedures for PC–RCUs running version 3.x.x are contained in TI 6130.6.

All other maintenance task procedures for PC–RCUs running version 4.x.x are contained in TI 6130.22.

5–51 – 5–74. Reserved.

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Section 3. SPECIAL MAINTENANCE PROCEDURES

5–75. General**.** All special maintenance procedures for the FDIO–G are contained in TI 6130.16.

All special maintenance procedures for PC–RCUs running version 3.x.x are contained in TI 6130.6.

All special maintenance procedures for PC–RCUs running version 4.x.x are contained in TI 6130.22.

5–76. – 5–99. Reserved.

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Appendix 1. Certification Requirements

Refer to Order 6000.15, for general guidance on the certification of systems, subsystems, and equipment.

NOTE: Service certification, if required, is the responsibility of the CONUS ARTCCs, ZAN and HCF facilities. Personnel at these facilities should refer to the latest version of Order 6100.1 and 6110.8 for requirements.

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Appendix 2. Glossary

Equipment, Peripheral	The auxiliary machines which may be placed under the control of a control unit. Examples of this are FDIO keyboards, monitors, printers.
Interface	The common boundary of two bodies or spaces. The functional intersystem relationship which influence system accomplishments.
Keyboard	A computer entry device used to enter messages into the computer for processing.
Message	A group of words transported as a unit.
National Airspace System	The common system of facilities, equipment regulations, procedures, and personnel required for the safe and efficient movement of civil and military aircraft in airspace within the jurisdiction of the United States.
Offline	In computer language, pertaining to auxiliary equipment or output device not under control of the central processing unit.
Online	In computer language, pertaining to operation of devices under direct control of the computer.
Operational Equipment	Equipment in actual use for air traffic control.
Output Equipment	Equipment used to transfer information out of a computer.
Software	In computer terminology, all the programming systems required for effective data processing operations.
Subsystem	An essential, functional part of a system which supports a data processing operation.

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Appendix 3. Acronyms

AFAAR	Aircraft Accident Representative
ARTCC	Air Route Traffic Control Center
ATC	Air Traffic Control
ATCT	Air Traffic Control Tower
ATO	Air Traffic Organization
BWM	Bandwidth Manager
CCD	Configuration Control Decision
CERAP	Combined Center Radar Approach Control
CONUS	Contiguous United States
CRT	Cathode Ray Tube
DoD	Department of Defense
DOT	Department of Transportation
EAS	En Route Automation System
ECG	En Route Communications Gateway
E&R	Exchange and Replace
FAA	Federal Aviation Administration
FBI	Federal Bureau of Investigation
FDIO	Flight Data Input/Output
FDIO-C	Flight Data Input/Output Center
FDIO-G	Flight Data Input/Output Gateway
FDIO-R	Flight Data Input/Output Remote
FDP	Flight Data Processor
FDR	Flight Data Router
FDTS	Flight Data Transfer System
FOIA	Freedom of Information Act
FOUO	For Official Use Only
FSEP	Facility, Services, and Equipment Profile
FTI	FAA Telecommunications Infrastructure
HCF	Honolulu Control Facility
HDD	Hardware Design Data
IAW	In Accordance With
IER	Name of Printer Manufacturer
ISP	Intelligent Switch Patch
ISSO	Information System Security Officer
LES	Law Enforcement Sensitive
LIS	Logistics Inventory System

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MEARTS	Micro En Route Automated Radar Tracking System
MTHB	Maintenance Technical Handbook
NAS	National Airspace System
NCP	NAS Change Proposal
NSA	National Security Agency
OCG	Offshore Flight Data Processing System (OFDPS) Communications Gateway
OFDPS	Offshore Flight Data Processing System
OUO	Official Use Only
Par.	Paragraph
PC	Personal Computer
PCII	Protected Critical Infrastructure Information
PC-RCU	Personal Computer–Remote Control Unit
POST	Power On Self-Test
RANK	Replacement Alphanumeric Keyboard
RAPCON	Radar Approach Control
RFSP	Replacement Flight Strip Printer
RSR	Remote Switching Router
S1R	Series /1 Replacement
SBU	Sensitive But Unclassified
SCSI	Small Computer System Interface
SHSI	Sensitive Homeland Security Information
SP	Support Processor
SSE	Servicing Security Element
SSI	Sensitive Security Information
SUI	Sensitive Unclassified Information
TD	Test Device
TFDM	Terminal Flight Data Manager
TOS	Technical Operations Services
TRACON	Terminal Radar Approach Control
VGA	Video Graphic Array
ZAN	Anchorage ARTCC
ZMA	Miami ARTCC

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Appendix 4. Policy/Procedures for Identifying, Handling, Marking and Disposal of Sensitive Unclassified Information (SUI)

1. Purpose. This appendix provides guidance to employees regarding the rules and practices for identification, handling, marking, storage and disposal of media that contains FOUO information, SSI, Sensitive Homeland Security Information (SHSI) and/or Protected Critical Infrastructure Information (PCII). This guidance does not pertain to any Classified (i.e., Confidential, Secret, and Top Secret) media.

This appendix provides guidance to employees regarding the rules and practices for identification, handling, marking, storage and disposal of media that contains FOUO information, SSI, Sensitive Homeland Security Information (SHSI) and/or Protected Critical Infrastructure Information (PCII). This guidance does not pertain to any Classified (i.e., Confidential, Secret, and Top Secret) media.

2. Scope. This guidance pertains to any Sensitive Unclassified Information (SUI) discussed verbally, transmitted electronically, existent or generated as printed material and/or contained on hard drives, disk drives, and/or any other type of computer storage media under the purview of the FAA. It pertains to every FAA employee, contractor, consultant and grantee creating, handling, or accessing SUI.

3. Background. In the aftermath of September 11, 2001, there is a heightened awareness of the need to safeguard sensitive Government information that does not meet the standards for classified national security information. Of particular concern is the need to protect Government information related to homeland security. This includes information that supports the FAA global aerospace structure that contributes to the security of the nation and public safety. It is incumbent upon all FAA associated personnel to support these safeguards through increased awareness of the nature of such material and their duties and responsibilities for its protection.

4. Policy. In accordance with (IAW) requirements stated in the latest edition of Order 1600.75, Protecting Sensitive Unclassified Information (SUI), SUI is unclassified information – in any form including print, electronic, visual, or aural forms - that must be protected from uncontrolled release to persons outside the FAA and indiscriminate dissemination within the FAA. It includes aviation security, homeland security, and protected critical infrastructure information. SUI may include information that may qualify for withholding from the public under the Freedom of Information Act (FOIA). All personnel shall be aware of and follow the policies and procedures contained herein and in the latest edition of Order 1600.75. If conflicting guidance is provided, comply with Order 1600.75.

5. Types of SUI. Throughout the Federal Government, there are more than 50 types of SUI. Within the FAA, only four types are generally encountered or handled:

a. FOUO information. FOUO is the primary designation given to SUI by the Department of Transportation (DOT) and FAA. It consists of information that could adversely affect the national interest, the conduct of Federal programs, or the privacy of individuals if released to unauthorized individuals. As examples, the uncontrolled use of FOUO information may allow someone to:

(1) Circumvent agency laws, regulations, legal standards, or security protective measures

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or

(2) Obtain unauthorized access to an information system**.**

b. SSI. SSI is a designation *unique* to the DOT and DOT's operating administrations and to the Department of Homeland Security (DHS). It applies to information we obtain or develop while conducting *security activities*, including research and development activities. Unauthorized disclosure of SSI would:

(1) Constitute an unwarranted invasion of privacy (including, but not limited to, information contained in any personnel, medical, or similar file);

(2) Reveal trade secrets or privileged or confidential information obtained from any person;

or

(3) Be detrimental to transportation safety or security.

c. SHSI. SHSI is a designation *unique* to homeland security information that we share with State and local personnel. The Federal government shares SHSI with State and local personnel who are involved in prevention against, preparation for, or response to terrorism. We protect it because its loss, misuse, unauthorized disclosure or access, or modification can significantly impair the capabilities and efforts of Federal, State, and local personnel to predict, analyze, investigate, deter, prevent, protect against, mitigate the effects of, or recover from acts of terrorism. If our sensitive unclassified information impairs these capabilities, we must designate it SHSI before we share it with State and local personnel to facilitate its proper protection.

d. PCII. PCII is a designation *unique* to critical infrastructure information provided by non-government persons and entities to the DHS. DHS uses the information for security of critical infrastructure and protected systems, analysis, warning, interdependency studies, recovery, reconstitution, or other informational purposes. While only DHS can designate information as PCII, they can share it with other Federal agencies as needed for operational purposes. PCII is defined in 6 CFR Part 29.

6. Identifying and Working with SUI. Personnel working with this system are most likely to encounter SUI identified above. However, other Federal agencies use different terminology and markings to designate SUI. For example, the Department of Energy uses Official Use Only (OUO), the Department of State uses SBU, and the Drug Enforcement Administration uses DEA Sensitive. Many Federal law enforcement agencies use the term Law Enforcement Sensitive (LES). If an unfamiliar security designation is encountered, contact the SSE for guidance; treat the information in question as FOUO until specific resolution is obtained.

Most existent SUI will be marked accordingly. However, marking is not conclusive proof of sensitivity. Information sensitivity may cease because of the passage of time or change in circumstances. Also, because of error or changing sensitivities, unmarked SUI that should be marked or information that is marked but is no longer sensitive may be encountered. If there is any uncertainty about markings, contact the supporting SSE for a resolution and protect the questioned information as though it were FOUO information pending resolution.

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The systems under cognizance of AJM-25, their operating systems and operational and backup data typically fall outside the scope of designation as SUI. However, the following specific information associated with AJM-25 programs and systems is considered SUI:

a. Lists and/or data files of employees and their respective system passwords or access codes. (FOUO)

b. System specific security information. (FOUO)

c. Procurement, grant, budget and other financial information and data. (FOUO)

d. Files and/or data bases containing specific private employee information such as medical histories, SSNs, personnel evaluations, etc. (SSI)

e. System or organizational vulnerability assessments. (SSI)

f. COOP documentation. (SSI)

g. SCAP documentation. (SSI)

h. Contingency planning documents, such as Disaster Recovery Plans. (SSI)

i. Trade secret or proprietary corporate information supplied in proposals and/or provided to the FAA under contract license agreements. (SSI)

A comprehensive, but generic listing of the specific information comprising SSI is provided in Order 1600.75, appendix A.

7. Marking/Labeling SUI. Marking is a basic protective measure that draws a reader's attention to the sensitivity of information and the need to protect it. Marking aids in making disclosure decisions and selecting and applying appropriate protective measures.

All personnel are required to properly mark information when created or it is determined that it meets the standards of sensitive unclassified information. Exceptions:

a. Records in storage. If there are unmarked records in storage that should be marked, they need not be removed from storage only to mark them. Mark them when removed from storage for other purposes other than destruction. Stored documents slated for destruction need not be marked prior to destruction provided the individual removing the document from storage can ensure destruction is accomplished as specified in paragraph 10 of this appendix.

b. Records marked under old authority. Information marked under an old regulatory authority, e.g., 14 CFR § 191 for Sensitive Security Information may be marked differently than directed by this order. Remarking these documents is not required.

c. Sensitive non-government information. Your office may receive information from contractors, grantees, businesses, and regulated parties marked as business sensitive, company confidential, proprietary, trade secret, and so on. Remarking this information is not required, but it must be protected from unauthorized disclosure. Unless greater protective measures are specified by the information's originator, protect it as FOUO information.

Refer to Order 1600.75, appendixes D and E for the most current guidance for marking FOUO information and SSI, which have different marking protocols. The SHSI Program Officer and PCII Officer will issue separate marking guidance for SHSI and PCII.

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When marking documents and other material containing both classified national security information and SUI, refer to FAA Order 1600.2, Classified National Security Information.

Mark other records, such as photographs, films, tapes, slides, or records residing in information systems with appropriate protective markings and distribution limitation statements in a conspicuous way so that persons having access to them are aware of their sensitivity.

Mark removable electronic media, e.g., diskettes and compact disks, with the appropriate protective marking and distribution limitation statement in a conspicuous way so that persons having access to them are aware of their sensitivity. The System ISSP will contain additional specific measures for labeling and marking removable media.

8. Storing SUI

a. During Working Hours

(1) Physical custody or control. When SUI is not in secure storage, it must be under the protection and control of an authorized person.

(2) *Not under* physical custody or control. When your SUI is not under the physical custody or control of an authorized person, you must store it in a lockable container, such as a file cabinet or desk, or in a locked space. Your office must control the keys to the locks of these containers and spaces, and key holders must be authorized persons.

b. After Working Hours

(1) Uncontrolled work spaces. If the work area is accessible to persons, who are not authorized access to the SUI, it must be stored in a secure container such as a locked desk, file cabinet, or an inaccessible locked space. The sub-team office must control the keys to the locks of these containers and spaces, and key holders must be authorized persons.

(2) Controlled work spaces. If the work area is accessible only to persons who are authorized access to the SUI, additional protective measures are not needed. Again, the sub-team office must control the keys to the locks for these controlled work spaces.

c. At Home or On Travel. Working on SUI outside the workplace poses additional security risks and challenges. If there is a need to work with SUI at home or while on travel, the system manager's approval to do so must be obtained. Each individual holder of SUI is responsible for protecting it from unauthorized disclosure while at home or traveling. Whether working and storing SUI from home or an assigned travel destination, the information is to be provided the same level of protection that it is afforded at the normal work site.

9. Distribution Procedures For SUI. SUI may be carried, mailed, or shipped it in any manner that prevents inadvertent disclosure of the contents. When sending SUI records outside the DOT, include supplementary markings and notices to explain the significance of the information and promote its proper handling. For example, include a statement such as the following in a transmittal record or directly on the record containing SUI:

This document/record belongs to the Federal Aviation Administration and may be used for official government purposes only. It may not be released without the express permission of the Federal Aviation Administration. Refer requests for the document to:

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(insert name and address of originating office.)

a. Hand carrying. Place the information in an opaque envelope or carry it within a brief case, pad folio, or other container. For FOUO information, use DOT Form 1600.7-1, FOUO Cover Sheet, or the FAA Form 1360-39, FOUO Envelope, if available.

b. Interoffice mail. Use a sealed opaque envelope with the addressee indicated on it. This is in addition to or instead of any office messenger envelopes. If available, use FAA Form 1360-39.

c. U.S. or Contract Mail. Mail or send documents and materials in properly addressed opaque envelopes or containers by United States Postal Service first-class, certified, or registered mail or contracted delivery service. Bulk shipments, such as directives, may be sent by 'fourth-class' mail provided the shipment is wrapped in opaque covering.

d. Telephone. Confirm that you are speaking to an authorized person before discussing the information, and inform the person that your discussion will include SUI and what part of the discussion is sensitive. Never leave voicemail messages containing SUI.

e. Fax.

(1) Mark the fax. Ensure that the documents being faxed are appropriately marked;

(2) Send to correct number. Use special care to ensure that documents are being sent to the correct fax number; and

(3) Determine how faxes are handled at the receiving end:

(a) If sending the fax to a controlled area, where only authorized persons will have access to it, then it may be sent without further precautions.

(b) If sending the fax to an uncontrolled area, where unauthorized persons might have access to it, then request an authorized person stand by at the receiving end while the fax is being sent. Ask for a confirming receipt.

f. Electronic Mail. Mark emails in the subject line "For Official Use Only" or "Sensitive Security Information" as appropriate. Send sensitive unclassified information as attachments; ensure they are appropriately marked IAW Appendices D and E of Order 1600.75. Also the security and encryption procedures of FAA Order 1370.121 must be followed.

g. Web Sites. Posting SUI to an unsecured web site that can be accessed by the public from the Internet is prohibited. Public web sites must not be provided links to web sites where SUI is posted. SUI may be posted to restricted FAA web sites provided they have special logon protocols and password protection. Passwords to these sites may be provided only to persons who satisfy the "duty to protect" and "need-to-know" requirements explained in Order 1600.75, chapter 2.

10. Disposal of SUI. The following guidance is provided for paper documents and record IAW requirements stated in Order 1600.75. Refer to the current version of this order to ensure this information is current.

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Destruction Standards for Paper Documents and Records		
If your document is:	Then your destruction standard is	Method
FOUO	To make recognition and reconstruction difficult	At a minimum, tearing or shredding each page into small pieces and mixing those pieces into regular trash
SSI	Completely to preclude recognition or reconstruction	Burning, shredding, wet-pulping and chemical decomposition (Note 1)
SHSI	By any means approved for destruction of classified information or by any other means that would make it difficult to recognize or reconstruct the information	Burning, cross-cut shredding, wet-pulping and chemical decomposition
Destruction Standards for Paper Documents and Records		
If your document is:	Then your destruction standard is	Method
PCII	By any method that prevents unauthorized retrieval	Burning, cross-cut shredding, wet-pulping and chemical decomposition
Note 1: Existing strip shredders may be used, but cross-cut shredding is preferred. Any new shredding equipment must have a cross-cut feature. The local Servicing Security Element (SSE) can provide assistance in selecting an appropriate destruction method and equipment.		

Refer to Appendix 5, Sensitive Security Information and Magnetic Media Disposal/Reutilization Policy/Procedures, for guidance pertaining to electronic/computer storage media.

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Appendix 5. Sensitive Security Information and Magnetic Media Disposal/Reutilization Policy/Procedures

1. Purpose. This appendix provides guidance to TOS employees who have a requirement to dispose of, or reuse, unclassified computer storage media that contains Sensitive Security Information (SSI) or For Official Use Only (FOUO) information. These procedures do not pertain to any Classified (i.e., Secret, Top Secret) media.

2. Scope. These procedures pertain to any Sensitive But Unclassified (SBU) data contained on hard drives, disk drives, and any other type of computer storage media under the purview of the FAA that contains SSI or FOUO information. Computer storage media located at FAA site(s), but owned or managed by other government agencies (i.e., Department of Defense (DoD), Federal Bureau of Investigation (FBI)), will not use these procedures, but will follow the procedures and/or guidelines for reusing or destroying magnetic media dictated by that particular agency. If the storage media is FAA-owned but contains sensitive information from another outside agency (i.e., DoD), then the sanitization procedures from the outside agency must be followed. If the outside agency's procedures are determined by the division ISSO to be better than or equal to the procedures outlined in this appendix, then no other action is necessary. If the outside agency's procedures are unknown or less stringent, then the procedures outlined in this appendix should be utilized.

3. Background. Computer storage media that contains SBU information cannot be reused or excessed unless all information has been destroyed or removed beyond comprehension. Reformatting the drive or deleting files will not completely remove the data from a system. There is still a great chance that the data can be recovered.

A number of methods can be used to accomplish proper sanitization. This document will provide three sets of procedures that will satisfy requirements necessary for reusing or disposing of electronic storage media. These procedures are:

a. overwriting the data manually and/or using approved data-overwriting software,

b. degaussing (i.e., demagnetizing the data), and

c. destroying the media.

4. Policy. In accordance with requirements stated in the latest edition of Order 1370.121, FAA Information Security and Privacy Program & Policy, to maintain the confidentiality and accountability of sensitive information, TOS FAA and support contractor personnel shall ensure confidentiality of sensitive data when reusing or disposing of magnetic media. The procedures contained in this document shall be followed when disposing of, or reusing, unclassified computer storage media that contains SSI and/or FOUO information. Authority to destroy magnetic media must be obtained from the owner or Property Custodian, whichever is appropriate, prior to executing any procedure that destroys data or the media.

All personnel shall follow the latest edition of Order 1600.2, Classified National Security Information, when disposing or reusing magnetic media that contains classified information.

5. Procedures for Sanitizing SBU Electronic Storage Media. Any one of the following procedures can be used when sanitizing any SBU electronic storage media before reuse or

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disposal. Before SBU media is to be removed or reused, the Division ISSO must be notified and consulted.

a. Overwriting Electronic Storage Media for Sanitization.

Overwriting is the process of replacing information (data) with meaningless data in such a way that the meaningful information cannot be recovered. The individual performing the overwriting will be responsible for certifying that the process has been successfully completed.

The process of overwriting data must be correctly understood and carefully implemented to be effective. Overwriting consists of recording data onto magnetic media by writing a pattern of binary ones (1) and zeros (0). These patterns can then be read back and interpreted as individual bits, eight (8) of which are used to represent a byte or character. If the data is properly overwritten with a pattern (e.g., 11111111" followed by 00000000") the magnetic fluxes will be physically changed and the drive's read/write heads will only detect the new pattern and the previous data will be effectively erased. To purge a hard drive requires overwriting with a pattern and then its complement, and finally with another pattern (e.g., overwrite first with 00110101," followed by 11001010" then 10010111"). Sanitization is not complete until all six passes of the three cycles are completed.

b. Data-overwriting Software.

Another method for sanitizing storage media is using commercial software. Certain software can be used but it must meet the criteria listed below, Paragraph 7, Acceptable Commercial Software Products, of this appendix. Software products and applications not meeting the stated minimum specifications are not acceptable for sanitizing SBU storage media. Overwriting software that reformats or repartitions a hard drive will not be accepted within the scope of this policy. Also, some software product versions may not have the capability to remove the Operating System (OS) during the overwrite process. To ensure the integrity of the sanitization process, overwriting software must have the following functions and capabilities:

(1) The ability to purge all data or information, including the OS, from the physical or virtual drives, thereby making it impossible to recover any meaningful data by keyboard or laboratory attack,

(2) A compatibility with, or capability to run independent of, the OS loaded on the hard drive,

(3) A compatibility with, or capability to run independent of, the type of hard drive being sanitized (e.g., ATA/IDE or Small Computer System Interface (SCSI) type hard drives),

(4) A capability to overwrite the entire hard disk drive independent of any Basic Input Output System (BIOS) or firmware capacity limitation that the system may have,

(5) A capability to overwrite using a minimum of three cycles (six passes) of data patterns on all sectors, blocks, tracks, and slack or unused disk space on the entire hard disk medium,

(6) A method to verify that all data has been removed from the entire hard drive and to review the overwrite pattern.

c. Degaussing.

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Degaussing is a procedure that reduces the magnetic flux of a medium to virtual zero by applying a reverse magnetizing field. Properly applied, degaussing renders any previously-stored data on magnetic media unreadable. The drawback to degaussing is that very seldom can a drive be used after this process. For a degausser to be effective here are a few standards and procedures which must be used:

(1) Degaussers used on FAA hard drives must have a nominal rating of at least 1700 Oersted.

(2) Degaussers must be operated at their full magnetic strength.

(3) The product manufacturer's directions must be carefully followed. Deviations from an approved method or rate of coercivity could leave significant portions of data remaining on a hard drive.

(4) All shielding materials (e.g., castings, cabinets, and mounting brackets) which may interfere with the degausser's magnetic field must be removed from the hard drive before degaussing.

(5) Hard disk platters must be in a horizontal direction during the degaussing process.

(6) For degaussing hard drives with very high coercivity ratings, it may be necessary to remove the magnetic platters from the hard drive's housing.

(7) Degaussing products should be acquired from the National Security Agency's (NSA) Degausser Products List which can be obtained by contacting:

National Security Agency

Attn: S7 Media Technology Center

9800 Savage Road, Ft. George G. Meade, MD

20755-6877

Tel: 1 (800) 688-6115 (Option #3) or 1 (410) 854-7661

Fax: 1 (410) 854-7668.

d. Destruction of Media.

Destruction of media is generally used when a disk has damaged or unusable tracks and sectors and the disk is not reusable. Authority to destroy the media must be obtained from the Property Custodian before proceeding. Destruction of storage media is the process of physically damaging a medium so that it is not usable in a computer, and so that no known exploitation method can retrieve data from it. If possible, operable media should be overwritten or degaussed prior to destruction. The three acceptable methods of destruction are:

(1) Physical destruction/impairment beyond reasonable use: Remove the hard drive from the chassis or cabinet. Remove any steel shielding materials, mounting brackets, and cut any electrical connection to the hard drive unit. In a suitable facility with individuals wearing appropriate safety equipment, subject the hard drive to physical force (i.e., pounding with a sledgehammer) that will disfigure, bend, mangle, or otherwise mutilate the hard drive so that it cannot be re-inserted into a functioning computer. Sufficient force should be used directly on

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top of the hard drive unit to cause shock/damage to the disk surfaces. In addition, any connectors that interface into the computer must be mangled, bent, or otherwise damaged to the point that the hard drive could not be re-connected without significant rework.

(2) Destruction at a metal destruction facility, (i.e., smelting, disintegration, or pulverization).

(3) Application of an abrasive substance (emery wheel or disk sander) to a magnetic disk or drum recording surface. Make certain that the entire recording surface is completely removed. Ensure proper safety measures, to include protection from inhaling abraded dust and use of protective eyewear.

6. Definitions.

a. Clearing – Rendering stored information unrecoverable unless special utility software or techniques are used.

b. Degaussing – Reduces the magnetic flux to virtual zero by applying a reverse magnetizing field. Properly applied, degaussing renders any previously stored data on magnetic media unreadable and may be used as a method of sanitization.

c. Media – Short for storage media. Physical objects on which data can be stored, such as hard drives, floppy disks, CD-ROMs, and tapes.

d. Overwriting – Process of writing patterns of data on top of the data stored on a magnetic medium.

e. Oerstad – A unit of magnetic field strength.

f. Sanitize - To expunge data from storage media so that data recovery is impossible. Sanitizing includes overwriting, degaussing, and destruction (destruction is not an appropriate means for TOS purposes). Clearing data does not constitute sanitizing.

g. SBU Information – SBU information is any information the loss, misuse, or unauthorized access to, or modification of, or the privacy to which individuals are entitled under Section 552a of Title 5, United Stated Code (The Privacy Act), but which has not been specifically authorized under criteria established by Executive Order or an Act of Congress to be kept secret in the interest of national defense or foreign policy. This includes information in routine FAA payroll, finance, logistics, and personnel management systems.

7. Acceptable Commercial Software Products.

The following commercial software is of acceptable use for overwriting computer storage media. This is not to be considered an all-inclusive list, as there are other products that meet the minimum requirements, but this list should be used as a reference. This list is subject to change and should only be used as a reference.

a. Product Name: No Trace™
Communication Technologies, Inc.,
14151 Newbrook Dr., Suite 400
Herndon, VA 20170
Tel: (703) 961-9080

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www.comtechnologies.com

b. Product Name: DataEraser™
ONTRACK Data International, Inc.
Tel: 1 (800) 872-2599
www.ontrack.com

c. Product Name: UniShred Pro™
Los Altos Technologies
Tel: (919) 233-9889
www.lat.com

d. Product Name: CleanDrive™
Access Data Corporation
(800) 574-5199
www.accessdata.com

e. Product Name: Sanitizer D 4.01™
Infraworks
(512) 583-5000
www.infraworks.com/products/sanitizer.html

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Appendix 6. Administration Information

1. Distribution. This directive is distributed to selected offices and facilities with the following Facility, Service, and Equipment Profile (FSEP) equipment: FDIOC and FDIOR.

a. To subscribe to email notifications of issued MTHBs:

(1) Go to http://dis.faa.gov. Click Subscribe. Click Subscribe to List. Enter your email address. Click Display Subscription Options. Select the series of MTHB for subscription. Click Subscribe.

or

(2) Go to https://employees.faa.gov/tools_resources/orders_notices. Select the series of MTHB for subscription. Click Go. Enter your email address. Click Go. Confirm email address, click radio button under Optional Password. Enter password if password was selected. Click Save.

b. For an electronic copy of this MTHB, go to https://employees.faa.gov/tools_resources/orders_notices/.

2. Authority to Change This Order. This MTHB can be modified or changed by Program Management Organization, En Route & Oceanic Second Level Engineering Group, AJM-25.

3. Definitions/Acronyms: Reference Appendix 2, Glossary, and Appendix 3, Acronyms.

4. Related Publications.

Order JO 1000.37	Air Traffic Organization Safety Management System
Order 1100.161	Air Traffic Safety Oversight
Order 1370.121	FAA Information Security and Privacy Program & Policy
Order 6000.15	General Maintenance Handbook for National Airspace System (NAS) Facilities
Order 6000.30	National Airspace System Maintenance Policy
Order 6000.50	National Airspace System (NAS) Integrated Risk Management
Order 6032.1	National Airspace System Modification Program
Order 1600.75	Protecting Sensitive Unclassified Information (SUI)
Order JO 6100.1	Maintenance of NAS En Route Stage A - Air Traffic Control System
Order JO 6110.8	Maintenance of the Offshore Flight Data Processing System (OFDPS)
Order JO 8020.16	Air Traffic Organization Aircraft Accident and Incident Notification, Investigation, and Reporting

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NAS-MD-311	Computer Program Functional Specification Message Entry and Checking Model A5f1.6
TI 6130.6	Terminal Flight Data Input/output (FDIO) System Hardware Design Data (HDD) Instruction Book: Personal Computer-Remote Control Unit (PC-RCU).
TI 6130.16	En Route Flight Data Input/Output (FDIO) System Hardware Design Data (HDD) Instruction Book: FGIO Gateway (FDIO-G)
TI 6130.17	En Route Flight Data Input/Output (FDIO) Instruction Book: FDIO Security Information
TI 6130.20	Offshore Flight Data Input/Output (FDIO) System Hardware Design Data (HDD) Instruction Book: Personal Computer–Remote Control Unit (PC–RCU) for Honolulu Control Facility (HCF) and Connected Remote Facilities
TI 6130.22	Terminal Flight Data Input/Output (FDIO) System Hardware Design Data (HDD) Instruction Book: Personal Computer–Remote Control Unit (PC–RCU) for sites with Flight Data Transfer Service (FDTS)

5. Forms and Reports. Forms and reports may be found at https://employees.faa.gov/tools_resources/forms/index.cfm

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Appendix 7. Safety Risk Management Documents

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